Lecture 14: Software Project Management

EGCI341: WEEK10

Outline

- Measures of Project Success
- Management Activities
- Project Plan Structure
- Activity Organization
- Project Scheduling
- PERT Chart
- Gantt Chart
- Activity Network
- Risk Management

Software Project Management

- Concerned with activities involved in ensuring that:
  - Software is delivered *on time* and *on schedule*

- Project management is needed because software development is always subject to:
  - *Budget* and *schedule* constraints that are set by the organisation developing the software

Software Management Distinctions

- Product is intangible (abstract)
- Product is uniquely flexible
- Software engineering is not recognized as an engineering discipline
- Software development process is *not standardised*
- Many software projects are 'one-off' projects (made only once)

Measures of Project Success

- The resulting information system is acceptable to the customer
- System was delivered "*on time*"
- System was delivered "*within budget*"
- System development process had a minimal impact on ongoing business operations

Project Staffs

- May not be possible to appoint the ideal people to work on a project
  - Project budget may not allow for the use of highly-paid staff
  - Staff with the appropriate experience may not be available

- Managers have to work within these constraints especially when there are shortages of trained staffs

Management Activities

- Proposal writing
- Project planning and scheduling
- Project costing
- Project monitoring and reviews
- Personnel selection and evaluation
- Report writing and presentations

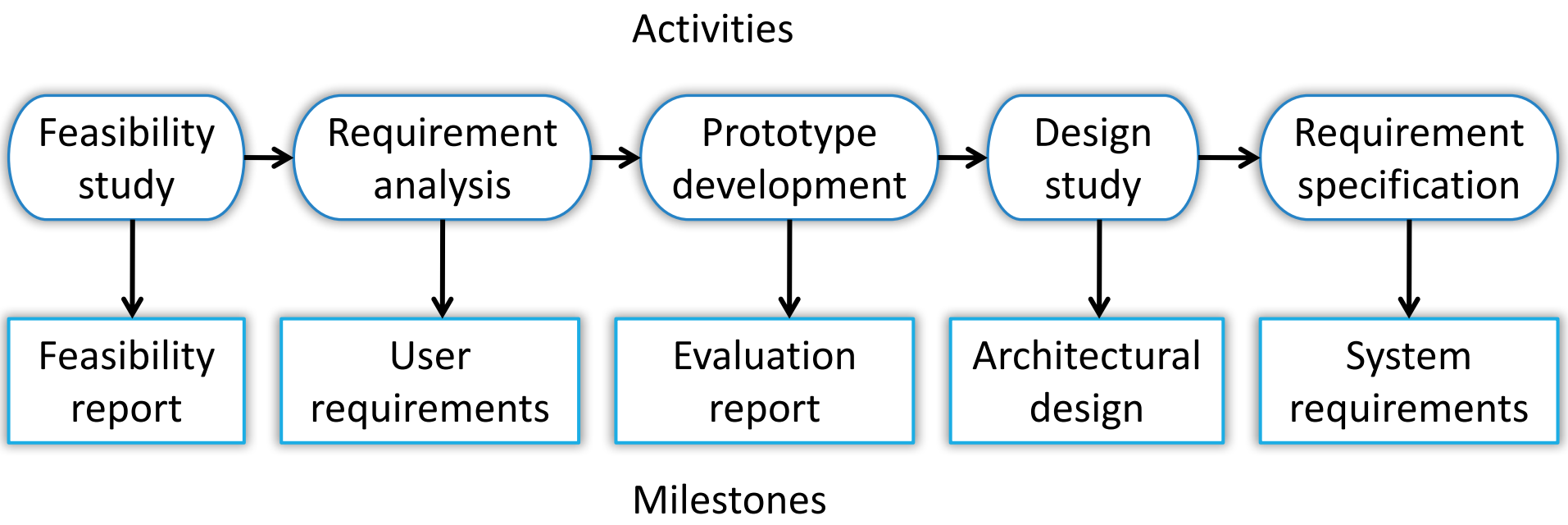
Project Plan

- Resources available to the project
- Work breakdown
- Schedule for the work

Activity Organization

- Activities in a project should be organised to produce tangible outputs for management to judge progress
- *Milestones* are the end-point of a process activity
- *Deliverables* are project results delivered to customers

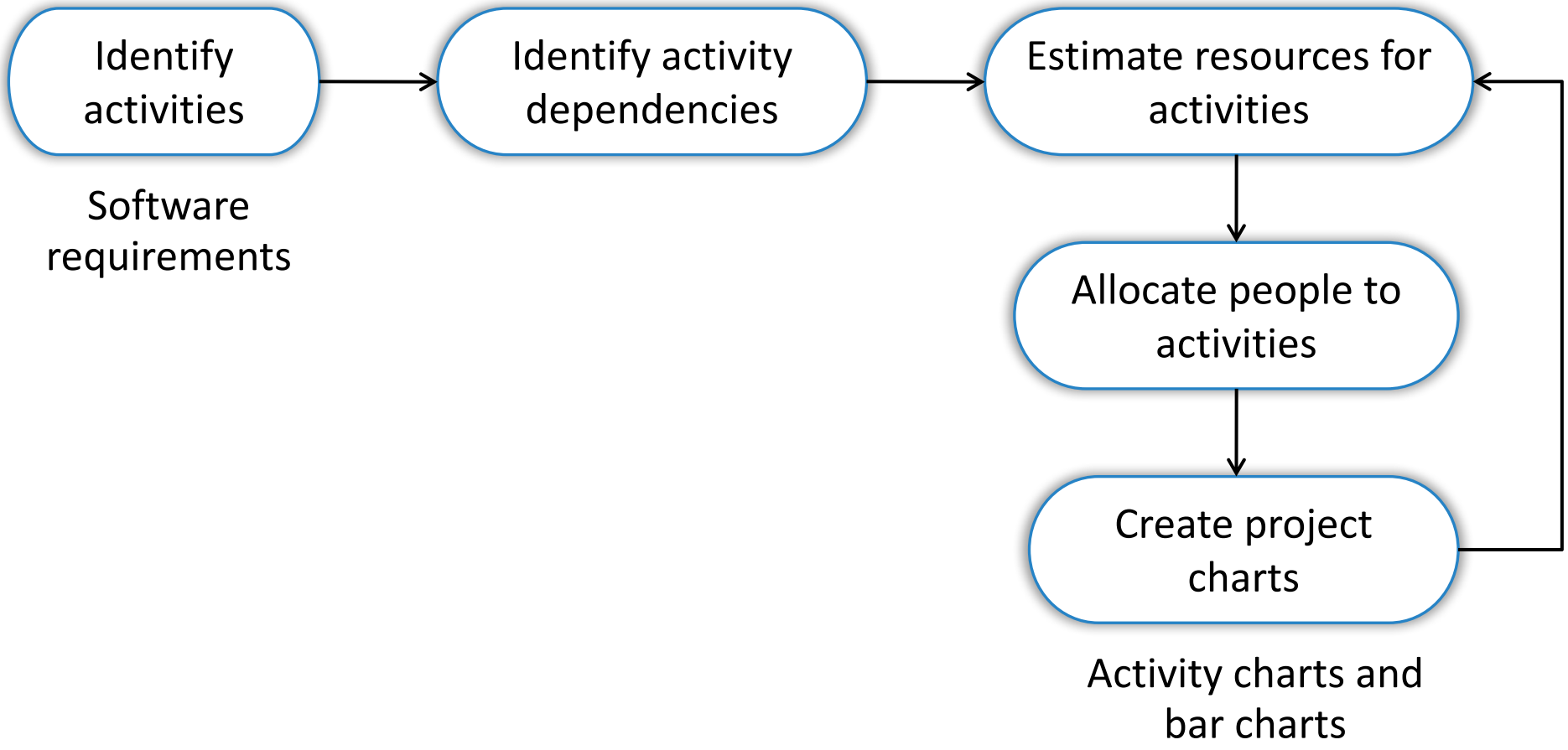
Activities & Milestones



Project Scheduling

- Split project into tasks and estimate time and resources required to complete each task
- Organize tasks concurrently to make optimal use of workforce
- Minimize task dependencies to avoid delays caused by one task waiting for another to complete
- Dependent on project managers intuition and experience

Project Scheduling Process



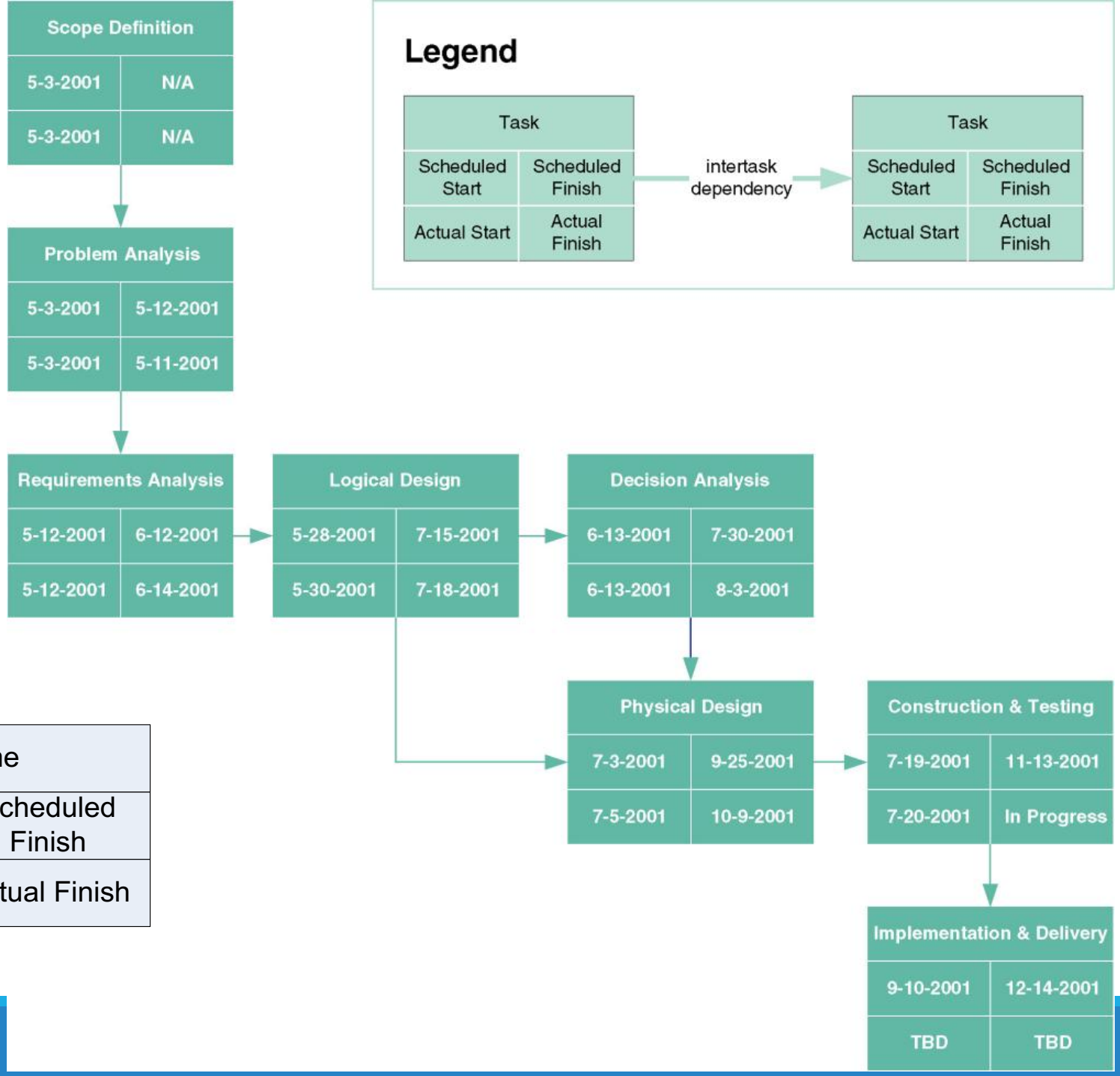
Scheduling Problems

- Estimating the difficulty of problems and hence the cost of developing a solution is hard
- Productivity is not proportional to the number of people working on a task
- Adding people to a late project makes it later because of communication overheads
- Unexpected always happens
  - Always allow contingency in planning

Bar Charts and Activity Networks

- Graphical notations used to illustrate the project schedule
- Show project breakdown into tasks
  - Tasks should not be too small
  - They should take about a week or two
- Activity charts show task dependencies and the critical path
- Bar charts show schedule against calendar time

PERT Chart



Task Name	
Scheduled Start	Scheduled Finish
Actual Start	Actual Finish

PERT Example

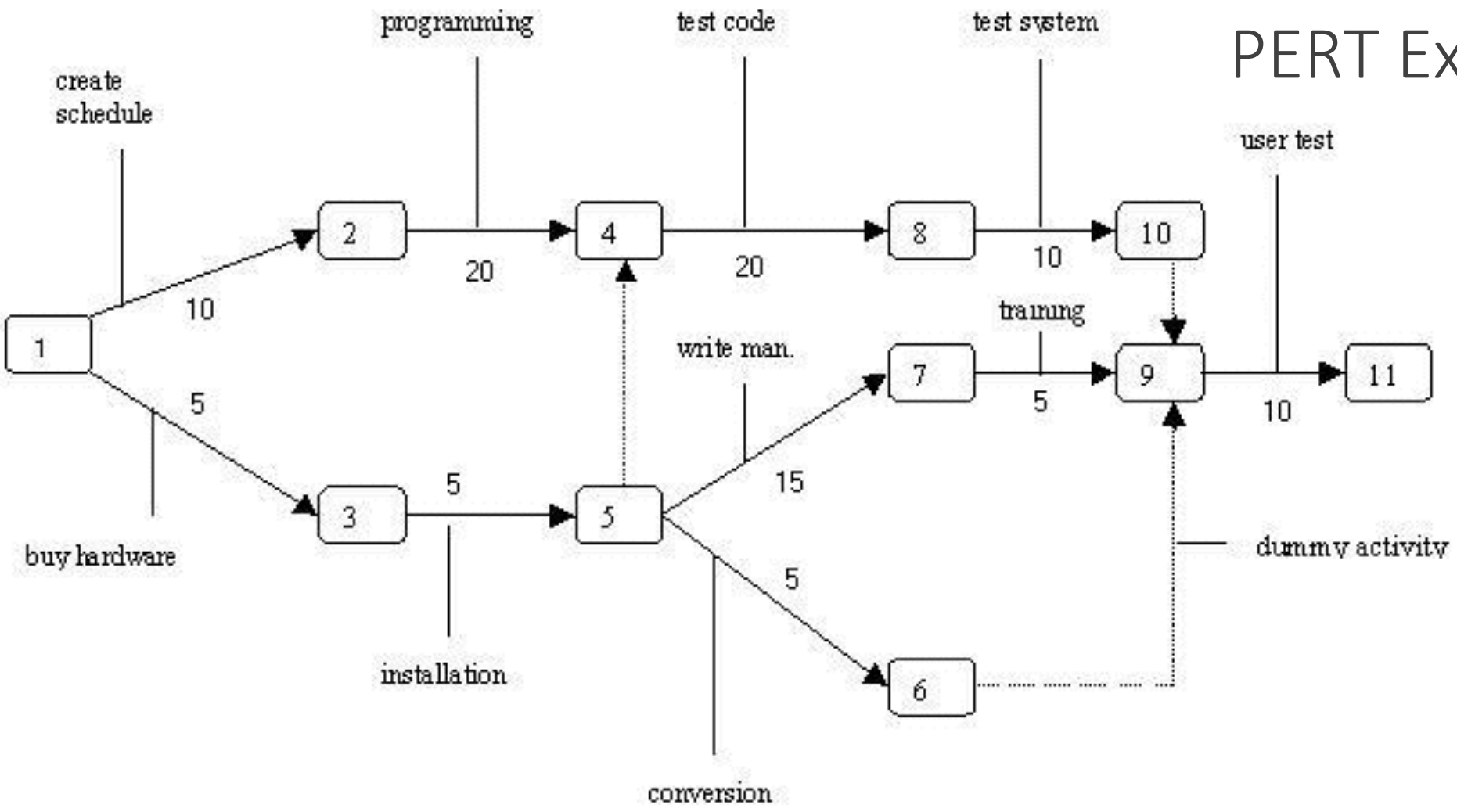
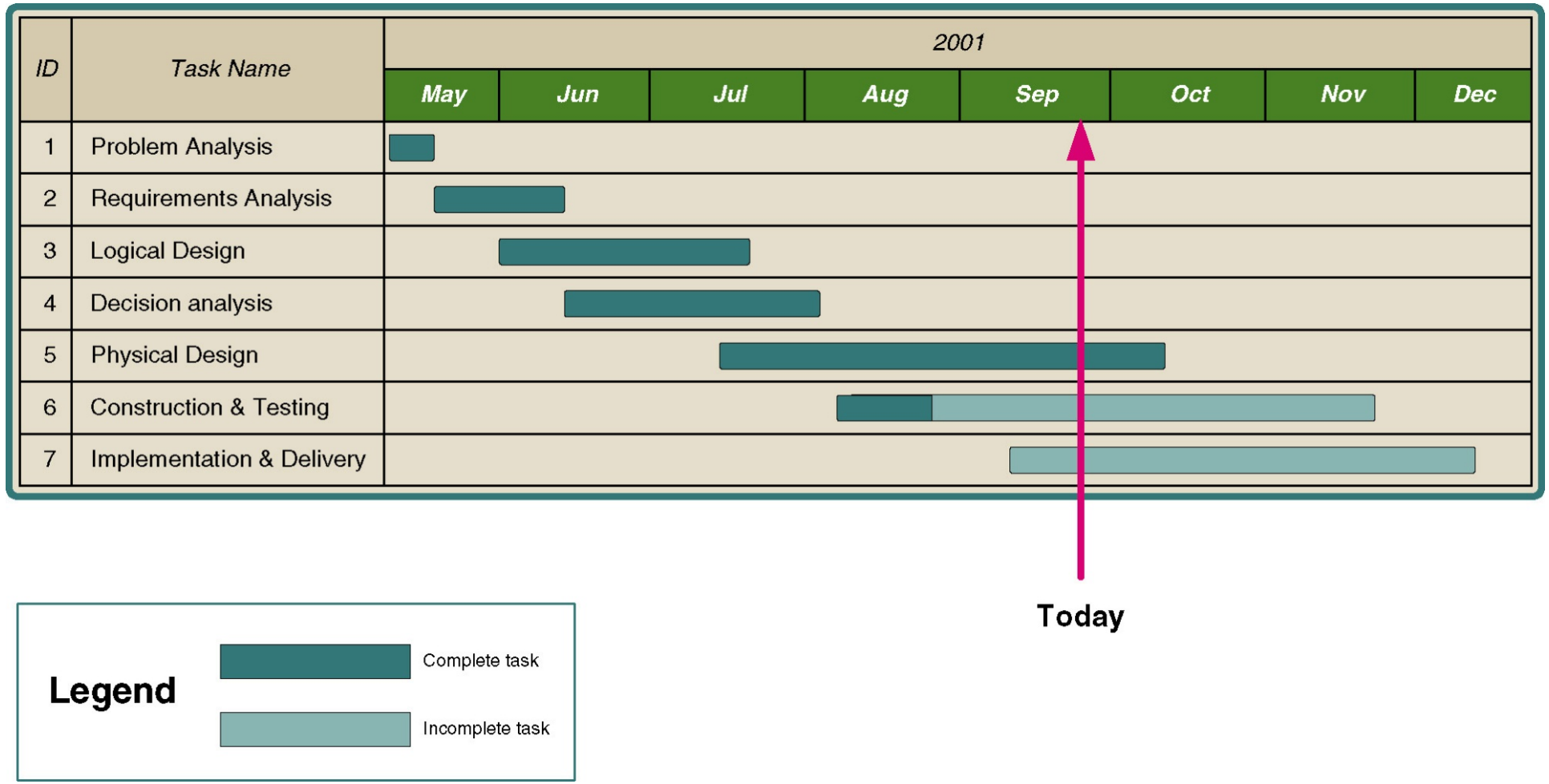


Fig. 1:
PERT Chart

* Numbered rectangles are nodes and represent events or milestones.
* Directional arrows represent dependent tasks that must be completed sequentially.
* Diverging arrow directions (e.g. 1-2 & 1-3) indicate possibly concurrent tasks
* Dotted lines indicate dependent tasks that do not require resources.

http://www.mckinnonsc.vic.edu.au/la/it/ipmnotes/ganttpert/pert_chart.gif

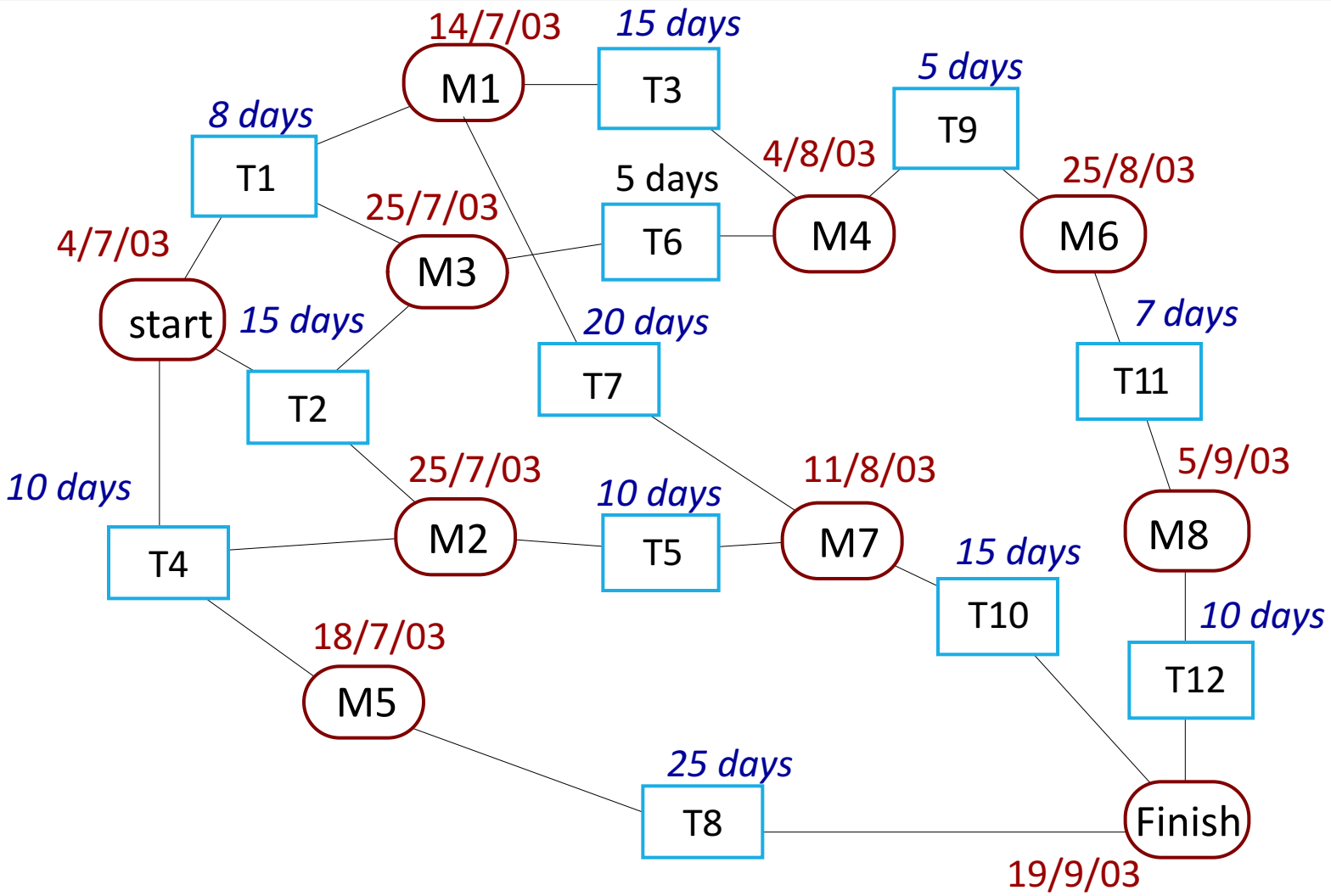
Gantt Chart


ID
Task Name
2001
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
1 Problem Analysis
2 Requirements Analysis
3 Logical Design
4 Decision analysis
5 Physical Design
6 Construction & Testing
7 Implementation & Delivery
Today
Legend
Complete task
Incomplete task

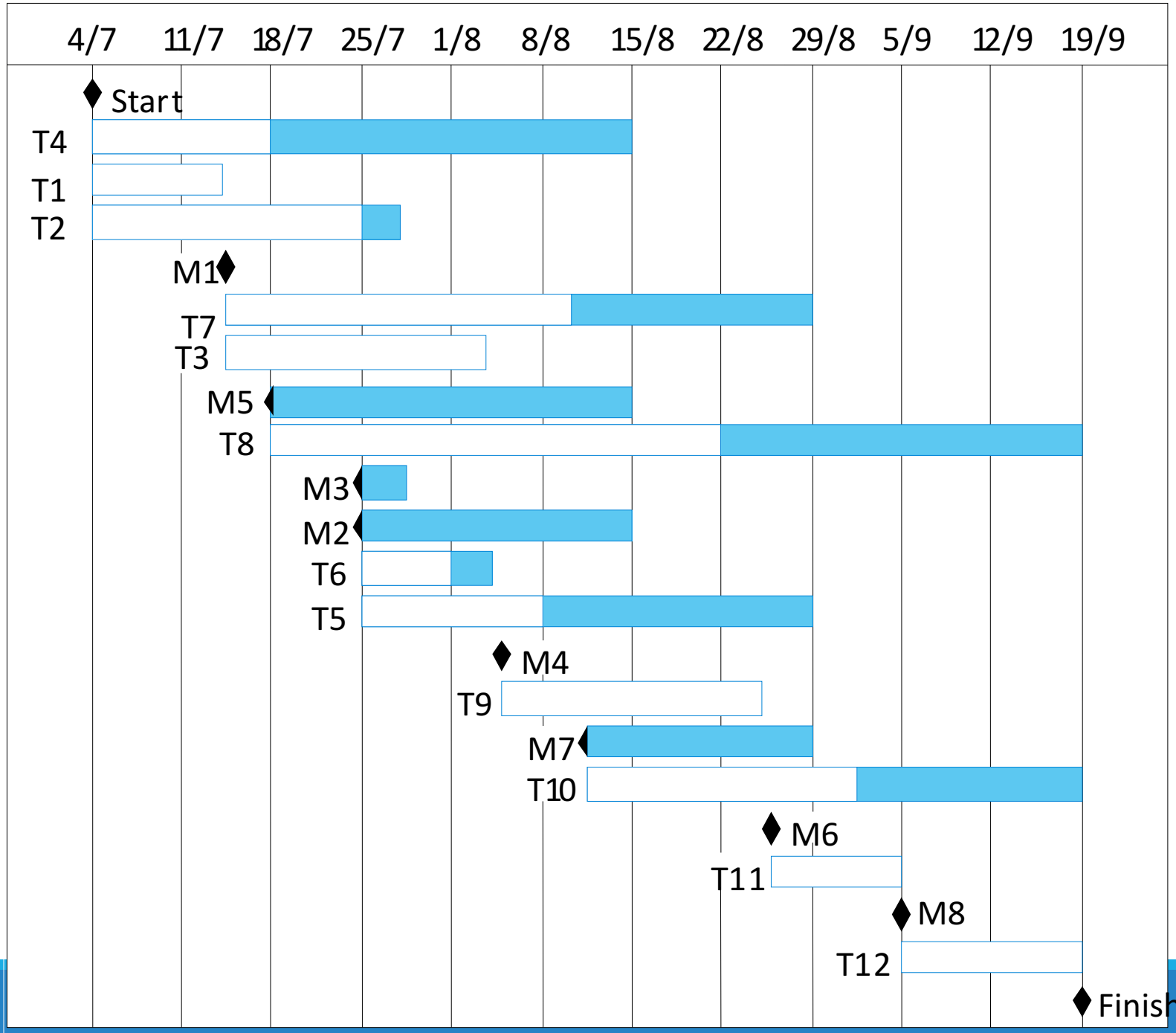
Task Durations and Dependencies

Activity	Duration (days)	Dependencies
T1	8	
T2	15	
T3	15	T1 (M1)
T4	10	
T5	10	T2, T4 (M2)
T6	5	T1, T2 (M3)
T7	20	T1 (M1)
T8	25	T4 (M5)
T9	15	T3, T6 (M4)
T10	15	T5, T7 (M7)
T11	7	T9 (M6)
T12	10	T11 (M8)

Activity Network



Activity Timeline



Estimate Task Durations

1. Estimate the minimum amount of time it would take to perform the task – the optimistic duration (OD)
2. Estimate the maximum amount of time it would take to perform the task – the pessimistic duration (PD)
3. Estimate the expected duration (ED) that will be needed to perform the task
4. Calculate a weighted average of the most likely duration (D) as follows:

$$D = \frac{(1 \times OD) + (4 \times ED) + (1 \times PD)}{6}$$

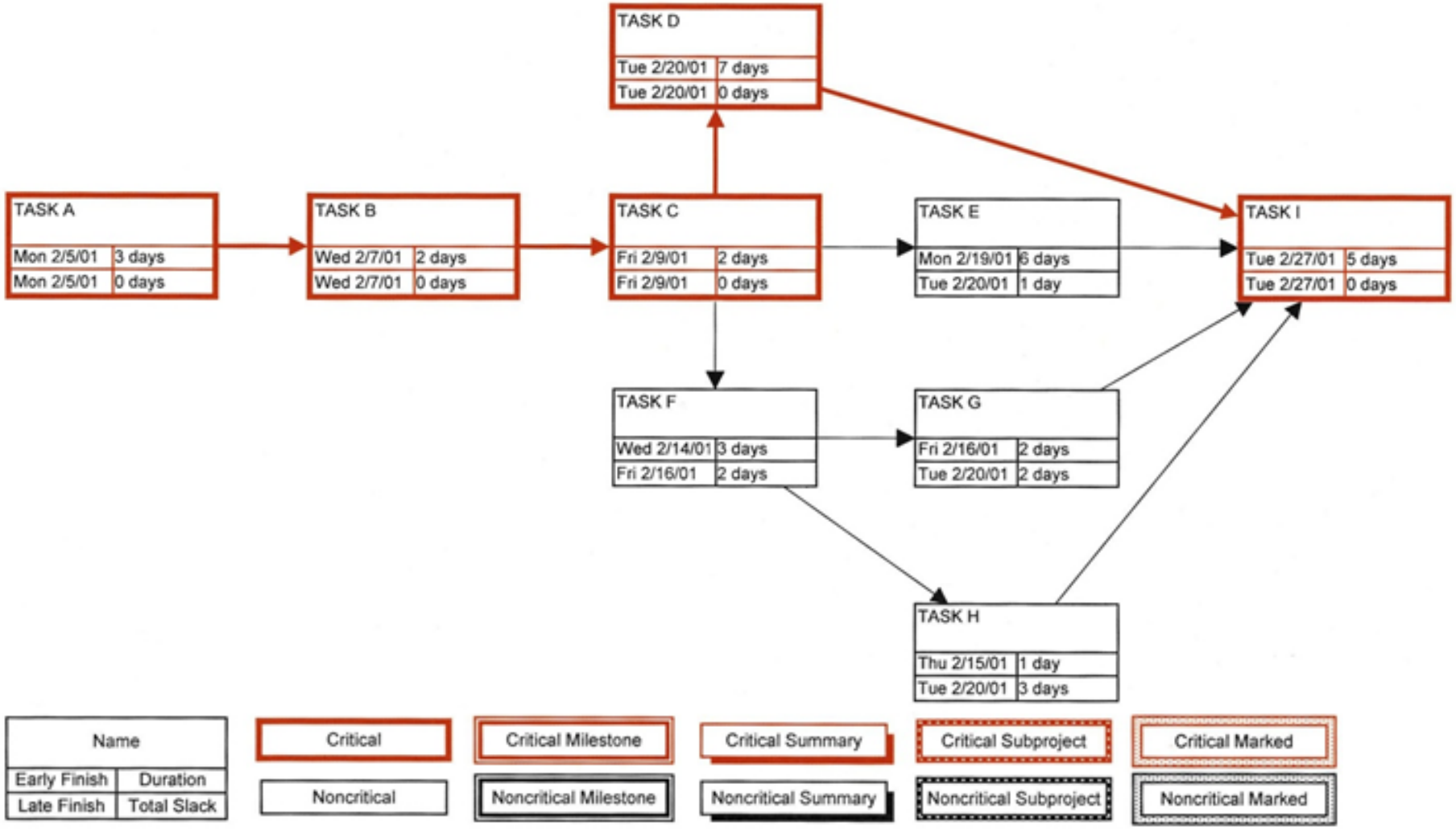
OD ED PD

$$3.33 \text{ days} = \frac{(1 \times 2 \text{ days}) + (4 \times 3 \text{ days}) + (1 \times 6 \text{ days})}{6}$$

Schedule Adjustments

- Using intertask dependencies, determine every possible path through the project
- Sum the durations of all tasks in each path
- **Path with the longest total duration** is the ***critical path***
  - **Critical Path** for a project is that sequence of dependent tasks that have *the largest sum of most likely durations*
    - Critical path determines the earliest completion date of the project
  - **Slack Time** available for any noncritical task is *the amount of delay that can be tolerated between the starting time and completion time of a task* without causing a delay in the completion date of the entire project

Critical Path Analysis



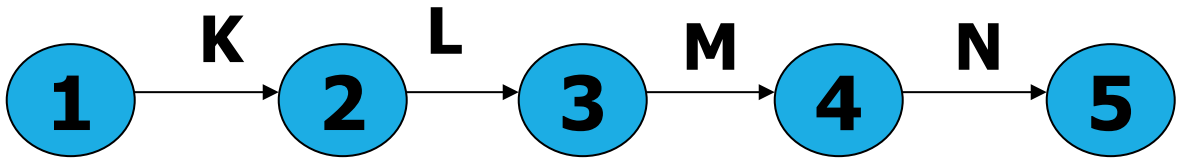
Example I

K – integration testing

L – Install Software

M – Write Manual

N – Train Users



Example II

Task	Days	Dependencies/ Predecessor
A	5	
B	3	
C	8	A, B
D	4	B
E	4	A
F	6	
G	4	C, D, E, F

Example III

- Draw Pert diagram for the scheduling table below
- Show the critical path and calculate duration of critical path
- Draw Gantt chart

Task	Description	Duration (Working Days)	Predecessor/s
A	Requirement Analysis	5	
B	Systems Design	15	A
C	Programming	25	B
D	telecoms	15	B
E	Hardware Installation	30	B
F	Integration	10	C, D
G	System Testing	10	E, F
H	Training/Support	5	G
I	Handover and Go-Live	5	H

Reference

- This set of slides and examples are modified from Ian Sommerville, Software Engineering 8th Edition, Addison-Wesley; 2007

Any Questions?

:O)

Thank you